Homework 2

Due Wednesday, 9 September 2026

1. The Weibull density function is given by

$$f(x;\alpha,m)=\begin{cases}\frac{1}{\alpha}mx^{m-1}e^{-x^m/\alpha} & x>0\\ 0 & \text{elsewhere}\end{cases}$$

where α and m are positive constants. The Weibull density function is often used as a model for the lifelengths of physical systems.

   a. Let Y follow an exponential distribution with mean parameter β. Prove that $W=\sqrt{Y}$ has a Weibull distribution with $\alpha=\beta$ and $m=2$. (Hint: this is a transformation problem)
   b. If a random sample of size n $(X_1,\dots,X_n)$ is taken from a Weibull distribution, find the pdf of the smallest order statistic $X_{(1)}$. What distribution, with parameters, does $X_{(1)}$ follow?

2. Let $X_1,\dots,X_n$ be a random sample from the Beta$(\alpha=2,\beta=2)$ distribution. Find the pdf of the largest order statistic, $X_{(n)}$.

3. Find a method-of-moments estimator (MME) of θ based on a random sample $X_1,\dots,X_n$ from each of the following distributions.

   a. $f(x;\theta)=\theta(1-\theta)^{x-1}$, $x=1,2,\dots;0\le\theta\le 1$
   b. $f(x;\theta)=(\theta+1)x^{-\theta-2}$, $x>1;\theta>0$
   c. $f(x;\theta)=\theta^2xe^{-\theta x}$, $x>0;\theta>0$

4. Let $X_1,\dots,X_n$ be a random sample from the pdf

$$f(x;\theta)=\theta x^{\theta-1},\qquad 0\le x\le 1;0<\theta<\infty$$

   Find a method-of-moments estimator of θ.